STA 325 Case Study

```
library(tidyverse)
library(knitr)

train_path <- "data/data-train.csv"
raw <- read_csv(train_path, na = c("", "NA"), show_col_types = FALSE)
```

Data Processing

We see that the existing data contains predictor values collected at different levels, instead of continuous values. It includes a valid infinity value for one of the predictors that is not a result of missing data. We compute standard deviation, skewness, and kurtosis from raw moments.

```
# predictor value distribution
level_table <- function(x, nm) {
  tibble(level = as.character(x)) |>
    count(level, name = "n") |>
    mutate(predictor = nm, .before = 1)
}
predictor_levels <- bind_rows(level_table(raw$Re, "Re"),
                              level_table(raw$Fr, "Fr"),
                              level_table(raw$St, "St"))
predictor_levels |>
  kable()
```

predictor	level	n
Re	224	39
Re	398	19
Re	90	31
Fr	0.052	36

predictor	level	n
Fr	0.3	23
Fr	Inf	30
St	0.05	6
St	0.1	6
St	0.2	7
St	0.3	7
St	0.4	6
St	0.5	6
St	0.6	5
St	0.7	7
St	0.8	7
St	0.9	8
St	1	5
St	1.5	6
St	2	7
St	3	6

```
# compute central moments
m1 <- raw$R_moment_1
m2 <- raw$R_moment_2
m3 <- raw$R_moment_3
m4 <- raw$R_moment_4

analysis <- raw |>
  mutate(
    variance = m2 - m1^2,
    mean = m1,
    sd = sqrt(variance),
    central_moment_3 = m3 - 3 * m2 * m1 + 2 * m1^3,
    central_moment_4 = m4 - 4 * m3 * m1 + 6 * m2 * m1^2 - 3 * m1^4,
    skewness = central_moment_3 / sd^3,
    kurtosis = central_moment_4 / sd^4
  )
```

```
response_names <- c("mean", "sd", "skewness", "kurtosis")
response_summary <- analysis |>
  summarise(across(all_of(response_names),
                   list(min = min, q1 = ~ quantile(.x, .25), median = median,
                        mean = mean, q3 = ~ quantile(.x, .75), max = max,
                        sd = sd))) |>
```

```
  pivot_longer(everything(), names_to = c("response", ".value"), names_pattern = "(.*)_(min|
response_summary |>
  kable(digits = 2)
```

response	min	q1	median	mean	q3	max	sd
mean	0.00	0.00	0.00	0.04	0.09	0.17	0.06
sd	0.01	0.16	0.28	3.65	0.73	32.32	8.95
skewness	11.97	72.55	110.12	162.81	269.54	344.91	113.10
kurtosis	150.53	5622.34	12158.72	39749.56	72732.45	132136.73	38100.12

Exploratory data analysis

```
predictor_dist <- analysis |>
  mutate(Fr_condition = if_else(is.infinite(Fr), "Inf", as.character(Fr))) |>
  group_by(Re, Fr_condition) |>
  summarise(n = n(),
            St_min = min(St),
            St_max = max(St),
            St_levels = n_distinct(St), .groups = "drop")
predictor_dist |>
  kable()
```

Re	Fr_condition	n	St_min	St_max	St_levels
90	0.052	12	0.05	3	12
90	0.3	10	0.10	3	10
90	Inf	9	0.05	3	9
224	0.052	14	0.05	3	14
224	0.3	13	0.05	2	13
224	Inf	12	0.05	3	12
398	0.052	10	0.10	3	10
398	Inf	9	0.05	2	9

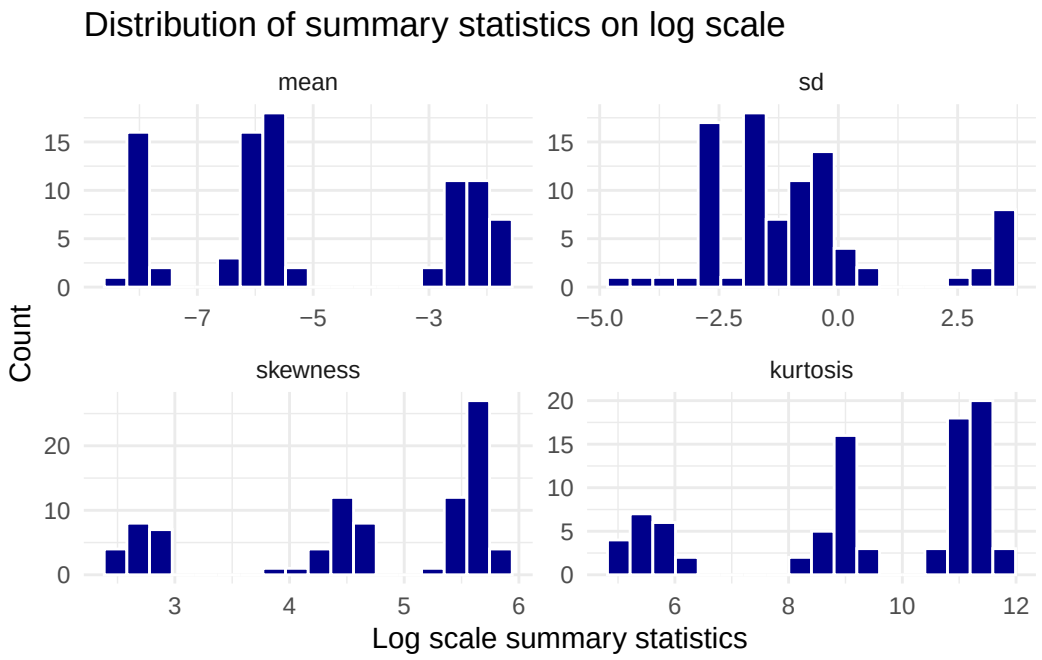
Response scales and associations within observed support

Histograms on both original and log scales address skew and candidate response transformations. The conditional plots assess whether the response changes with `St`, whether curvature is plausible, and whether observed trends differ by `Re` or `Fr`. Log axes are used only for strictly

positive responses. Smooths are descriptive LOESS fits within the observed `Re`–`Fr` facets, not inferential tests.

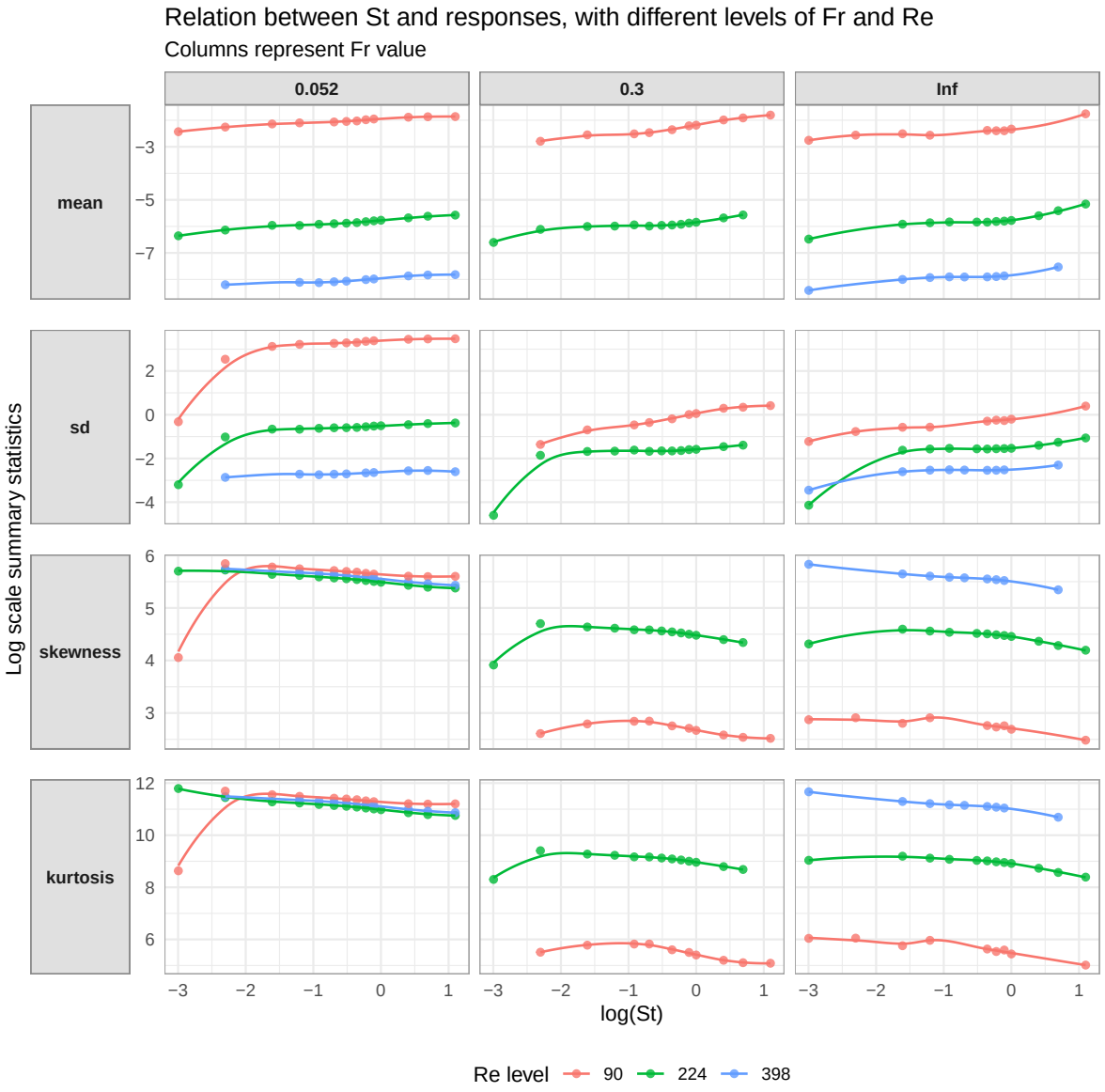
```
response_long <- analysis |>
  select(Re, Fr, St, all_of(response_names)) |>
  pivot_longer(all_of(response_names), names_to = "response", values_to = "value") |>
  mutate(Fr_condition = if_else(is.infinite(Fr), "Inf", as.character(Fr)),
         log_value = log(value),
         log_St = log(St),
         response = fct_relevel(response, "mean", "sd", "skewness", "kurtosis"))

ggplot(response_long, aes(log_value)) +
  geom_histogram(bins = 18, color = "white", fill = "darkblue") +
  facet_wrap(~ response, scales = "free") +
  labs(x = "Log scale summary statistics",
       y = "Count",
       title = "Distribution of summary statistics on log scale") +
  theme_minimal()
```




```
ggplot(response_long, aes(log_St, log_value, color = factor(Re))) +
  geom_point(alpha = .8) +
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, linewidth = .6) +
  facet_grid(rows = vars(response), cols = vars(Fr_condition), scales = "free_y",
```

```
            switch = "y") +
labs(x = "log(St)",
     y = "Log scale summary statistics",
     color = "Re level",
     title = "Relation between St and responses, with different levels of Fr and Re",
     subtitle = "Columns represent Fr value") +
theme_minimal() +
theme(
  panel.border = element_rect(color = "grey65", fill = NA, linewidth = .45),
  panel.spacing.y = grid::unit(1.1, "lines"),
  strip.placement = "outside",
  strip.background = element_rect(fill = "grey88", color = "grey55"),
  strip.text.y.left = element_text(face = "bold", angle = 0),
  strip.text.x = element_text(face = "bold"),
  legend.position = "bottom"
)
```


Relation between St and responses, with different levels of Fr and Re
Columns represent Fr value
0.052
0.3
Inf
mean
sd
skewness
kurtosis
Log scale summary statistics
log(St)
Re level
90
224
398